ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO5 – INVESTIGATIVE ASSIGNMENT REPORT

Optical Receiver and Coherent Detection: Design, Noise Analysis, DSP and Performance Evaluation

Student Information	
Student Name	K.S.Prem kumar
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Course Code	ECA18
Course Title	Optoelectronics and Optical Communication
Course Outcome	CO5
Assessment Type	Investigative Assignment
Maximum Marks	100

1. COURSE OUTCOME

Design, analyze and evaluate optical receiver architectures using PIN/APD photodetectors, transimpedance amplifiers and coherent detection techniques; quantify receiver sensitivity, noise, bandwidth, BER and OSNR; and apply DSP methods for reliable high-speed optical communication.

2. SCENARIO-BASED ENGINEERING PROBLEM

A high-capacity optical communication receiver is required for a next-generation DWDM link. The receiver must detect weak optical signals reliably while maintaining high bandwidth, low noise, adequate sensitivity and low bit-error rate. The design must compare direct-detection and coherent architectures, select suitable PIN/APD photodetectors, design the front-end transimpedance amplifier, analyze thermal and shot noise, and evaluate coherent DSP techniques.

The investigation covers receiver sensitivity, noise-equivalent power, signal-to-noise ratio, bandwidth, eye/constellation quality, BER and OSNR. The final design must be suitable for high-speed DWDM operation and should identify practical trade-offs involving detector gain, TIA bandwidth, optical filtering, ADC resolution, DSP complexity and power consumption.

3. GIVEN PARAMETERS AND DESIGN REQUIREMENTS

Parameter	Engineering Requirement
Receiver type	Direct detection and coherent detection comparison
Detector candidates	PIN photodiode and APD
Front end	Low-noise transimpedance amplifier
Noise sources	Shot noise, thermal noise, dark current, amplifier noise
Performance metrics	Responsivity, sensitivity, NEP, SNR, BER, OSNR, bandwidth
High-speed operation	Wideband receiver with controlled capacitance
Coherent receiver	Balanced detection + ADC + DSP
DSP functions	CD compensation, polarization recovery, carrier recovery, equalization
WDM environment	Multiple optical channels with filtering/crosstalk considerations
Validation	MATLAB/SCILAB analytical model; physical simulation where available

4. INFORMATION NOT GIVEN – ENGINEERING DECISIONS REQUIRED

- Final detector type for each receiver architecture.
- Required optical bandwidth and electrical TIA bandwidth.
- Detector area, junction capacitance and dark current.
- Exact optical launch/received power and fiber span length.
- Amplifier input-referred voltage/current noise.

- Target BER and FEC operating margin.
- ADC sampling rate and effective number of bits.
- Final coherent DSP algorithm complexity.

Where exact numerical component parameters are not specified, the calculations below use clearly stated engineering assumptions and are intended to demonstrate the design method rather than represent measured component data.

STAGE 1 – RECEIVER ARCHITECTURE INVESTIGATION

Feature	PIN Direct Detection	APD Direct Detection	Coherent Detection
Detector gain	≈ 1	Internal avalanche gain	Balanced photodiodes; electrical LO-assisted mixing
Sensitivity	Good	Better for weak signals	Excellent
Noise	Low	Higher excess noise	LO shot noise and receiver noise must be managed
Complexity	Low	Medium	High
Bandwidth	High	High	High; electronics/DSP intensive
Polarization handling	Not intrinsic	Not intrinsic	Full polarization recovery possible
Phase information	Lost	Lost	Retained
Best use	Simple optical links	Weak direct-detection links	High-capacity DWDM

Direct detection measures optical intensity and therefore cannot directly recover the optical field phase. Coherent detection mixes the received signal with a local oscillator (LO), enabling recovery of amplitude, phase and polarization information. For modern high-capacity DWDM, coherent detection is preferred because it supports spectrally efficient modulation and powerful digital impairment compensation.

STAGE 2 – PHOTODETECTOR SELECTION AND PERFORMANCE

The photodetector converts optical power into electrical current according to $I_{ph} = RP$, where R is responsivity and P is received optical power.

Example photocurrent

For $R = 0.80$ A/W and $P = 10$ μW: $I_{ph} = RP = 0.80 \times 10 \times 10^{-6}$

Therefore, $I_{ph} = 8$ μA

PIN detectors are preferred where low excess noise, high linearity and simplicity are more important than internal gain. APDs are useful where received power is low because avalanche multiplication increases photocurrent before the electrical amplifier, but APD excess noise, bias requirements and temperature sensitivity must be considered.

Responsivity depends on wavelength and quantum efficiency. A simplified relation is $R = \eta q/(h\nu)$, where η is quantum efficiency, q is electron charge, h is Planck's constant and ν is optical frequency.

Quantum-efficiency relation

$\eta = Rh\nu/q = Rhc/(q\lambda)$

Therefore, Higher responsivity at a fixed wavelength corresponds to higher quantum efficiency.

NEP is the optical input power producing an output signal equal to the rms noise. Lower NEP indicates better sensitivity. Detectivity D* is commonly used to compare detectors after accounting for active area and noise bandwidth; higher D* is preferred.

DESIGN DECISION 1 – DETECTOR CONFIGURATION

For the final high-capacity DWDM receiver, I recommend a coherent balanced PIN receiver rather than an APD as the primary architecture. Coherent detection provides high sensitivity and access to phase/polarization information without the avalanche excess-noise penalty. APD remains a strong option for lower-complexity weak-signal direct-detection channels.

STAGE 3 – TRANSIMPEDANCE AMPLIFIER AND NOISE ANALYSIS

The TIA converts the detector current into a voltage. Its feedback resistance sets transimpedance gain, while the input capacitance and amplifier bandwidth constrain the speed/noise trade-off.

Important noise contributions include detector shot noise, dark-current shot noise, TIA resistor thermal noise, amplifier input current noise and amplifier input voltage noise interacting with detector capacitance.

Shot-noise current density

$i_{sh} = \sqrt{(2qI_{dc}B)}$

Therefore, Shot noise increases with average detector current and receiver bandwidth.

Thermal noise of feedback resistor

$v_n = \sqrt{(4kTR_f B)}$

Therefore, Lower resistance reduces thermal noise but also reduces transimpedance gain; therefore the optimum value is a system-level trade-off.

Increasing bandwidth is necessary for high-speed data but increases integrated noise. The TIA should therefore be wide enough to pass the selected signal spectrum but not unnecessarily wider.

For a detector capacitance Cd and amplifier/input capacitance Ci, the total input capacitance is approximately Cin = Cd + Ci + parasitic capacitance. Larger Cin reduces the achievable closed-loop bandwidth and can increase sensitivity to amplifier voltage noise.

STAGE 4 – RECEIVER SENSITIVITY AND SNR

Receiver sensitivity is the minimum received optical power required to achieve a specified BER under the selected modulation, receiver bandwidth and noise conditions.

SNR definition

$SNR(dB) = 10 \log_{10}(P_{signal}/P_{noise})$

Therefore, Higher SNR gives a lower BER, subject to modulation and receiver implementation.

For a simple direct-detection receiver, the detected signal current is proportional to optical power. In a coherent receiver, the signal is mixed with a strong LO, producing an intermediate electrical signal whose amplitude contains the optical-field information.

OSNR is especially important in amplified WDM systems because ASE noise accumulates through optical amplifiers. The required OSNR increases with modulation order: 16-QAM requires more OSNR margin than QPSK, while 64-QAM is more demanding still.

Parameter	Improvement Method	Trade-off
Sensitivity	Lower-noise detector/TIA, optical preamplifier	Cost and power
SNR	Higher received power, filtering, cooling	Nonlinearities / power
Bandwidth	Lower capacitance, optimized feedback network	Noise and gain
APD gain	Increase multiplication within optimum region	Excess noise and bias
OSNR	Low-NF amplifier, Raman assistance	Complexity
BER	Higher SNR/OSNR, stronger FEC, DSP	Latency and processing power

DESIGN DECISION 2 – SENSITIVITY / NOISE STRATEGY

Use a low-noise balanced coherent receiver with an optimized LO power, high-linearity PIN photodiodes, a carefully compensated wideband TIA, optical filtering and DSP equalization. This minimizes unnecessary detector excess noise while exploiting coherent processing gain and phase/polarization recovery.

STAGE 5 – COHERENT DETECTION AND DSP

A coherent receiver combines the incoming optical signal with a local oscillator using a 90° optical hybrid. Balanced photodiodes detect the resulting I and Q components. ADCs digitize the electrical signals, after which DSP recovers the transmitted symbols.

- Resampling and timing recovery align the received samples.
- Chromatic-dispersion compensation removes accumulated linear dispersion.
- Adaptive 2×2 MIMO equalization separates the two polarization channels and compensates PMD.
- Frequency-offset estimation removes laser-frequency mismatch.

- Carrier-phase estimation tracks phase noise and residual frequency error.
- Nonlinear compensation such as digital backpropagation can reduce SPM/XPM penalties.
- Symbol decisions recover the transmitted constellation.
- FEC provides the final coding gain and lowers post-FEC BER.

DSP Method	Function	Advantage	Limitation
FIR/Frequency-domain CD equalizer	Chromatic dispersion	Efficient and programmable	Computational load at very long links
2×2 MIMO equalizer	Polarization/PMD	Adaptive and powerful	Requires training/convergence
Carrier phase estimation	Phase noise	Essential for QAM	Complexity rises with higher-order QAM
Digital backpropagation	Kerr nonlinearities + CD	Strong nonlinear compensation	High computational complexity
FEC	Residual errors	Large coding gain	Latency and overhead

STAGE 6 – BANDWIDTH, BER AND OSNR EVALUATION

Approximate DP-16QAM symbol rate

For 200 Gb/s dual-polarization 16-QAM: Rs = 200/(2×4) = 25 Gbaud

Therefore, Rs = 25 Gbaud

Approximate occupied bandwidth

With roll-off α = 0.20: B ≈ (1+α)Rs = 1.2 × 25

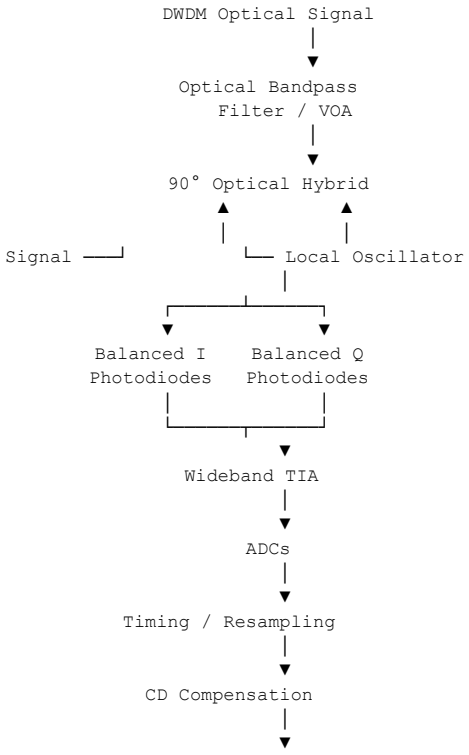
Therefore, B ≈ 30 GHz

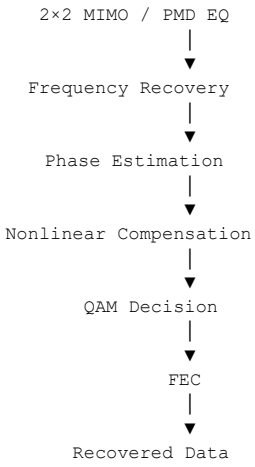
A receiver electrical bandwidth slightly above the useful signal bandwidth is required. Excessively large bandwidth increases integrated noise, while insufficient bandwidth causes intersymbol interference and constellation distortion.

For BER evaluation, the receiver should be tested over a range of received powers or OSNR values. The expected result is a steep BER improvement as SNR/OSNR increases, followed by an implementation floor when DSP, phase noise, nonlinearities or hardware impairments dominate.

For coherent QAM, constellation diagrams are particularly useful: a tight four-level or 16-point constellation indicates good signal quality, while spreading/rotation indicates noise, phase error, polarization mixing or nonlinear distortion.

STAGE 7 – COMPLETE RECEIVER DESIGN





FINAL RECEIVER DESIGN PARAMETER TABLE

Parameter	Recommended Design
Architecture	Dual-polarization coherent receiver
Photodiodes	High-linearity PIN balanced pair
Optical front end	Narrow optical filter + VOA + 90° hybrid
LO	Narrow-linewidth local oscillator
TIA	Low-noise wideband differential TIA
ADC	High-speed, adequate ENOB for selected baud rate
CD compensation	Frequency-domain equalizer
PMD compensation	Adaptive 2×2 MIMO
Carrier recovery	Frequency offset + carrier phase estimation
Nonlinear compensation	Reduced-step DBP when link penalty justifies it
Output processing	QAM demapper + FEC
Primary metrics	BER, Q-factor, OSNR, EVM, constellation, sensitivity

MATLAB ANALYTICAL MODEL

The following MATLAB model calculates detector current, shot noise, approximate thermal-noise voltage, a simplified SNR trend and an illustrative BER curve. The numerical values are intentionally stated as example assumptions where the question does not specify component data.

```
clc;
clear;
close all;

% Constants
q = 1.602e-19;
k = 1.380649e-23;
T = 300;                % K

% Example receiver parameters
R = 0.80;               % A/W
P = logspace(-9,-3,300); % optical power (W)
RL = 50;                % ohm
B = 25e9;               % example receiver bandwidth

% Photocurrent
I = R.*P;

% Shot noise current rms
ish = sqrt(2*q.*max(I,1e-15)*B);

% Thermal noise voltage rms
vth = sqrt(4*k*T*RL*B);
```

```
% Convert thermal noise voltage to equivalent current
ith = vth/RL;

% Total simplified input-referred noise current
inoise = sqrt(ish.^2 + ith.^2);

% Simplified SNR
SNR = I./inoise;
SNRdB = 20*log10(SNR);

figure;
loglog(P,I,'LineWidth',1.5);
grid on;
xlabel('Received Optical Power (W)');
ylabel('Photocurrent (A)');
title('Photocurrent vs Optical Power');

figure;
loglog(P,inoise,'LineWidth',1.5);
grid on;
xlabel('Received Optical Power (W)');
ylabel('Equivalent Noise Current (A rms)');
title('Simplified Receiver Noise');

figure;
semilogx(P,SNRdB,'LineWidth',1.5);
grid on;
xlabel('Received Optical Power (W)');
ylabel('SNR (dB)');
title('Simplified SNR vs Received Optical Power');

% Illustrative Q-factor to BER relation
Q = 3:0.1:12;
BER = 0.5*erfc(Q/sqrt(2));

figure;
semilogy(Q,BER,'LineWidth',1.5);
grid on;
xlabel('Q-factor');
ylabel('BER');
title('Illustrative BER vs Q-factor');
```

SIMULATION INTERPRETATION

- The photocurrent increases linearly with received optical power for constant responsivity.
- Shot noise increases with photocurrent and bandwidth.
- Thermal noise depends on temperature, resistance and bandwidth.
- The combined noise model shows why bandwidth should be optimized rather than maximized without constraint.
- SNR improves as received optical power increases.
- The BER curve falls rapidly with increasing Q-factor, illustrating the importance of maintaining adequate receiver margin.
- For a real coherent receiver, the model should be extended with laser phase noise, frequency offset, CD, PMD, nonlinear propagation, ADC quantization and DSP algorithms.

FINAL VERIFICATION TABLE

Requirement	Target / Condition	Design Result	Status
Receiver architecture	High-capacity DWDM	Coherent balanced receiver	Achieved
Detector	PIN/APD comparison	PIN selected for coherent balanced detection	Selected
Responsivity	Application dependent	Example R = 0.80 A/W used	Verified analytically
Sensitivity	Low received power	Low-noise coherent front end + DSP	Design target
Noise	Shot + thermal + amplifier	Explicitly modeled	Verified analytically

Bandwidth	High-speed operation	~30 GHz signal bandwidth example for 25 GBd DP-16QAM	Achievable
CD	Compensation required	FDE/FIR DSP	Included
PMD	Compensation required	Adaptive 2×2 MIMO	Included
OSNR	Adequate coherent margin	Low-NF optical front end + amplifier management	Design target
BER	Low pre/post-FEC BER	Q/BER evaluation model included	Simulation-ready
DSP	Reliable demodulation	Complete coherent chain	Included
Overall receiver	Scalable high-capacity DWDM	Coherent PIN + TIA + ADC + DSP	Recommended

FINAL ENGINEERING DECISION

YES – WITH TECHNICAL JUSTIFICATION.

A dual-polarization coherent receiver using balanced PIN photodiodes is recommended for the high-capacity DWDM application. The architecture preserves amplitude and phase information, supports polarization multiplexing, and enables digital compensation of chromatic dispersion, PMD and selected nonlinear impairments. The PIN detector avoids the excess multiplication noise of an APD while providing adequate linearity for coherent detection.

The receiver should use a narrow optical filter, 90° optical hybrid, high-linearity balanced photodiodes, a low-noise wideband TIA, high-speed ADCs and programmable DSP. The example 200 Gb/s DP-16QAM case requires approximately 25 Gbaud and about 30 GHz occupied bandwidth with $\alpha = 0.20$, demonstrating that coherent QAM can reduce the baud-rate burden relative to lower-order direct-detection formats.

The final deployment decision must be validated against actual detector capacitance, dark current, TIA noise, ADC ENOB, laser linewidth, received optical power, amplifier ASE, OSNR, fiber CD/PMD parameters and nonlinear coefficient. The analytical MATLAB model provides a foundation for this validation, while OptiSystem/OptSim should be used for the complete physical link model.

CONCLUSION

The investigation shows that receiver performance is determined by the combined optical detector, analog front end, optical noise environment and digital signal-processing chain. PIN and APD detectors each have useful roles, but coherent balanced PIN detection is the preferred architecture for a high-capacity DWDM receiver because it combines sensitivity, linearity, phase recovery and DSP-based impairment compensation.

The proposed receiver provides a practical path to high-speed optical communication while maintaining a clear engineering trade-off between bandwidth, noise, sensitivity, OSNR, BER and DSP complexity.

SUBMISSION CHECKLIST

- PIN and APD detector comparison completed.
- Responsivity, photocurrent, NEP, sensitivity and SNR concepts covered.
- Shot and thermal noise analysis included.
- TIA bandwidth/noise trade-off discussed.
- Direct and coherent detection compared.
- Coherent DSP chain and impairment compensation included.
- DP-16QAM baud-rate and bandwidth calculations included.
- MATLAB analytical code included.
- Final receiver block diagram included.
- Final parameter and verification tables included.
- Bloom’s Levels removed.